

RESEARCH ARTICLE

Genomic characterization of the dominating Beta, V2 variant carrying vaccinated (Oxford–AstraZeneca) and nonvaccinated COVID-19 patient samples in Bangladesh: A metagenomics and whole-genome approach

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Abstract

Bangladesh is experiencing a second wave of COVID-19 since March 2021, despite the nationwide vaccination drive with ChAdOx1 (Oxford–AstraZeneca) vaccine from early February 2021. Here, we characterized 19 nasopharyngeal swab (NPS) samples from COVID-19 suspect patients using genomic and metagenomic approaches. Screening for SARS-CoV-2 by reverse transcriptase polymerase chain reaction and metagenomic sequencing revealed 17 samples of COVID-19 positive (vaccinated = 10, nonvaccinated = 7) and 2 samples of COVID-19 negative. We did not find any significant correlation between associated factors including vaccination status, age or sex of the patients, diversity or abundance of the coinfecting organisms/pathogens, and the abundance of SARS-CoV-2. Though the first wave of the pandemic was dominated by clade 20B, Beta, V2 (South African variant) dominated the second wave (January 2021 to May 2021), while the third wave (May 2021 to September 2021) was responsible for Delta variants of the epidemic in Bangladesh including both vaccinated and unvaccinated infections. Noteworthy, the receptor binding domain (RBD) region of S protein of all the isolates harbored similar substitutions including K417N, E484K, and N501Y that signify the Beta, while D614G, D215G, D80A, A67V, L18F, and A701V substitutions were commonly found in the non-RBD region of Spike proteins. ORF7b and ORF3a genes underwent a positive selection (dN/dS ratio 1.77 and 1.24, respectively), while the overall S protein of the Bangladeshi SARS-CoV-2 isolates underwent negative selection pressure (dN/dS = 0.621). Furthermore, we found different bacterial coinfections like *Streptococcus agalactiae*, *Neisseria meningitidis*, *Elizabethkingia anophelis*, *Stenotrophomonas maltophilia*, *Klebsiella pneumoniae*, and *Pseudomonas plecoglossicida*, expressing a number of antibiotic resistance genes such as *tetA* and *tetM*. Overall, this approach provides

valuable insights on the SARS-CoV-2 genomes and microbiome composition from both vaccinated and nonvaccinated patients in Bangladesh.

KEYWORDS

antimicrobial resistance gene (AMR), coinfection, metatranscriptomics (mRNA), SARS-CoV-2, variants

1 | INTRODUCTION

Since the emergence of the COVID-19 pandemic, it continues to cause substantial mortality and plunge the global economies into deep recessions.¹ As of December 21, SARS-CoV-2 has resulted in more than 271 million confirmed cases worldwide, with over 5.32 million fatalities (<https://www.worldometers.info/coronavirus/>). Although a global vaccination program is underway, the emergence of novel lineages harboring a cluster of mutations in the RBD (receptor binding domain) and immunodominant regions of N-terminal domain of spike glycoprotein, evolving immune escape variants with increased transmissibility and infectivity.² Studies revealed that it is likely to escape from protective antibodies elicited by natural infection or vaccination. In some cases, it exhibits a complete immune escape in convalescent serum samples, which leads to rising concern regarding the effectiveness and efficacy of the existing vaccines.²⁻⁴

In Bangladesh, the nationwide mass vaccination drive was started with Oxford–AstraZeneca on February 9. As of December 2021, more than 140 million doses of currently approved vaccines (e.g. Oxford–AstraZeneca, Moderna, Pfizer/BioNTech, Johnson & Johnson, Sinopharm, Sinovac) have been administered (<https://covid19.trackvaccines.org/country/bangladesh/>; <https://covidvax.live/location/bgd>). In the midst of a vaccination campaign, Bangladesh saw the second wave of COVID-19 in March 2021, using the UK variation, as the starting point (Alpha, V1). However, during the third week of March 2021, the Beta, V2 variant from South Africa has taken over as the most predominant variant, which replaces the UK variant and others,⁵ thus leading to the resurgence of COVID-19 infection throughout the country with increased death and hospitalization (<https://www.thedailystar.net/coronavirus-deadly-new-threat/news/south-african-variant-covid-19-dominantbangladesh-icddr-study-2073757>).

Furthermore, in the mid of May 2021, the third wave of COVID-19 emerged by Delta variants and proved to be the deadliest in Bangladesh, and has led to a dramatic surge in COVID-19 cases and deaths domestically.⁶ This variant of concern delta was first identified in India with the evidence of possessing higher transmissibility, and alarmingly more severe disease outcomes (e.g., increased rate of hospitalizations and deaths).⁷ In the meantime, a new variant of concern omicron (B.1.1.529) has been reported from South Africa in November and rapidly transmitted in at least 57 countries of the world as of December 9.⁸⁻¹⁰ Till now, two cases of omicron have been reported from Bangladesh.¹⁰ Recent studies suggested that the accumulation of complicated mutations in the spike glycoprotein may lead this variant to

escape from the protective immune response elicited by vaccination or natural infection.⁹ Moreover, several studies have already elucidated that these variants of concern carrying mutations on spikes showed reduced susceptibility to convalescent-phase serum and plasma samples from COVID-19 patients.¹¹ However, the beta variant gained serious attention due to its significantly reduced level of neutralization efficacies to currently available vaccines such as Moderna, Pfizer–BioNTech, Oxford–AstraZeneca, Novavax, etc.¹² Further studies demonstrated the least efficacies of Oxford–AstraZeneca vaccine against the beta variant, which exhibits only 22% efficacy against mild to moderate cases involving beta variant.¹³ That is far below the 50% minimum standards for emergency use.¹⁴ Thus, the recent emergence of multiple immune escape variants of SARS-CoV-2 has disrupted confidence about whether the current generation of vaccines will be able to provide long-term protection against infection.^{4,15}

Progression and the extent of disease outcome of COVID-19 seem to be dependent on multiple factors comprising host-specific risk factors (e.g. age, sex, etc.), comorbidities (e.g. cardiovascular disease, diabetes, chronic respiratory disease, cancer, etc.), and multiple intertwined hosts' pathophysiological mechanisms.¹⁶ The heterogeneity of disease course observed among the patients is not solely attributed to the viral infection but also to the differential hosts' immune responses.¹⁷ However, administration of immunomodulatory interventions to reduce the severe outcomes (active inflammatory responses and cytokines storms) may lead to inefficient clearance of virus, while introducing an avenue for a plethora of microbial communities including virus, bacteria, archaea, fungus, etc. to evolve and interact together along with SARS-CoV-2.¹⁸ Worryingly, the coexistence of other microbial pathogens and underlying intricate molecular pathogenesis leading to exorbitant indisposition in severely ill COVID-19 patients.¹⁹ In this scenario, empiric broad-spectrum use of antibiotics has been prescribed for the majority of the COVID-19 patients irrespective of the cause. Unfortunately, heavy antimicrobial administration may result in a significant increase in antimicrobial resistance (AMR) and antifungal resistance development and dissemination, attributing increased mortality owing to the limited arsenal of antimicrobial to treat community-acquired infections.²⁰

Metagenomic next-generation sequencing (mNGS) and whole-genome sequencing is a recently developed high-throughput approach, facilitates to sequence the entire genetic content present in a particular specimen. This technology allows to detect and describe a wide range of known and unknown emerging pathogens, including their virulence and/or AMR profiles.¹⁸ This study thus was carried

out to precisely analyze the meta-transcriptomic sequencing data to reveal the genomic and metagenomic features among vaccinated and nonvaccinated populations, though the complete host–pathogen interaction may not be elucidated.

2 | METHODOLOGY

2.1 | Sample collection and screening for SARS-CoV-2

A total of 19 nasopharyngeal (NP) swab samples (male = 13, female = 6) have been selected for RNA extraction according to their C_T values. SARS-CoV-2 has been primarily detected by using SANSURE RT-PCR kit (SANSURE BIOTECH INC. Chana) as per the manufacturer's protocol where we have assigned the patients samples into the following three criteria: 10 patients were vaccinated (first dose, Oxford–AstraZeneca); 7 patients were nonvaccinated; 2 patients were suspected having COVID-19 but negative by laboratory reverse transcriptase polymerase chain reaction (RT-PCR) test. Information about all the studied samples as mentioned in Table 1 covers the patient's age, sex, vaccination status, overall C_T values, and genomic accession numbers.

2.2 | Nucleic acid extraction and mRNA sequencing

Whole-genome sequencing of all SARS-CoV-2 isolates was conducted by the Genomic Research Laboratory of BCSIR, Bangladesh. Viral RNA was extracted using ReliaPrep™ Viral Total Nucleic Acid Purification Kit (catalog no. AX4820). Viral cDNA and DNA libraries were prepared using Illumina RNA Prep with Enrichment (L) Tagmentation (Illumina, catalog no. 1000000124435). After tagmentation and amplification, samples were enriched with the Respiratory Virus Oligos Panel V2 (Illumina, catalog no. 1000000124435) and library pool the sample was sequenced using the MiniSeq™ sequencing System with an output of paired-end (2×74 bp) reads. Sequenced reads of SARS-CoV-2 genomes were assembled using Basespace Drogen RNA Pathogen Detection software version 3.5.14 with default settings.

2.3 | SARS-CoV-2 genome assembly and quality control

After sequencing, SARS-CoV-2 genome reads were detected in the RT-PCR positive samples ($n = 17$) only (Table 1). The sequencing technology produces an average of 1.05 million (maximum = 5.01 million, minimum = 0.07 million) reads per sample. After trimming and quality control, the samples are subject to SARS-CoV-2 consensus generation with EzCOVID19 (<https://www.ezbiocloud.net/tools/sc2>). Of EZBioCloud. The average coverage of the SARS-CoV-2

genome was $\times 2943$ (maximum = $\times 7623$, minimum = $\times 78$) (Table 1). After consensus generation, the genomes are subject to further quality check using snippy²¹ to observe area-wise coverage of each variation position. Trimmomatic v0.39,²² Burrows–Wheeler Aligner (BWA v0.7.17),²³ SAMtools v1.12,²⁴ and BEDTools v2.30.0²⁵ were used for area-wise sequence arresting for final quality check. The error-corrected genomes were then used for downstream analyses.

2.4 | Phylogenetic reconstruction and selection pressure analysis

Nextstrain SARS-CoV-2 phylogenetic tree generation pipeline (<https://nextstrain.org/sars-cov-2>) has been used to reconstruct the phylogenetic tree with the nextstrain clade of the sequenced ($n = 17$) and Bangladeshi sequences ($n = 2174$) retrieved from GISAID with having complete and high coverage complete genome published till December 13, 2021.²⁶ Nextstrain uses MAFFT²⁷ as an alignment tool, IQ-TREE²⁸ and TreeTime²⁹ to reconstruct tree and Auspice web server (<https://auspice.us/>) to visualize the tree. Then the selection pressure of all complete Bangladeshi sequences ($n = 1025$) was calculated by the SLAC algorithm³⁰ in the Datamonkey server (<http://www.datamonkey.org/>) with a p -value threshold of ≤ 0.1 to understand their evolutionary pattern. Datamonkey uses Hyphy (Hypothesis Testing using Phylogenies) to calculate the direction of selection.

2.5 | Mutation analysis and variant determination of SARS-CoV-2 genome

Mutations and clades have been assigned from the CoVserver mutation app (<https://www.gisaid.org/epiflu-applications/covserver-mutations-app/>) of Global initiative on sharing all influenza data (GISAID) where hCoV-19/Wuhan/WIV04 (GenBank accession no. MN996528.1) was used as a reference genome. PANGO Lineage of the genomes has been assigned from Pangolin COVID-19 Lineage Assigner (<https://pangolin.cog-uk.io/>). Nextstrain²⁶ clade of those same genomes has also been assigned with Nextclade v0.14.2 (beta) (<https://clades.nextstrain.org/>).

2.6 | Shotgun metatranscriptomic (mRNA) experimental approaches of studied samples

Once again, the COVID-19 pandemic accentuates the necessity to consider coinfections and secondary infections during and after the viral outbreaks, respectively. To find other microbial coinfections, AMR defense along with SARS-CoV-2 infection from Bangladeshi people, this study further analyzed sequenced data (previously mentioned) against nucleotide database (nt) of National Center for Biotechnology Information (NCBI), ARG-Annot-2 database. A total of 17 RT-PCR-positive and 2 RT-PCR-negative samples were considered for this coinfection searching study.

TABLE 1 Suspected COVID-19 sample characteristics and sequencing results

Sample ID	RT-PCR C _T value	SRA accession no.	GISAID accession no.	Age	Sex	Vaccination	Raw read count	Average coverage (x)	Assembly length (bp)	Clade	Lineage ^a	Nextstrain clade
DU-BCSIR-17	24	SRR14484350	EPI_ISL_2037865	54	Male	Yes	356,162	186.66	29,835	GH	B.1.351	Beta.V2
DU-BCSIR-18	25	SRR14484349	EPI_ISL_2038623	17	Male	No	106,879	124.48	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-19	26	SRR14484348	EPI_ISL_2038944	37	Male	Yes	2,111,057	6803.98	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-20	23	SRR14484347	EPI_ISL_2039394	55	Female	Yes	272,7819	7140.44	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-21	25	SRR14484346	EPI_ISL_2039757	40	Male	Yes	5,012,511	7623.07	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-22	25	SRR14484345	EPI_ISL_2040202	44	Male	No	553,478	2421.26	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-23	24	SRR14484361	EPI_ISL_2043212	25	Female	No	279,369	1055.29	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-24	24	SRR14484360	EPI_ISL_2043671	60	Male	No	336,912	951.07	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-25	23	SRR14484359	EPI_ISL_2044075	19	Male	No	2,379,772	7054.55	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-26	24	SRR14484358	EPI_ISL_2044525	54	Female	Yes	2,367,234	6850.73	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-27	23	SRR14484357	EPI_ISL_2044950	65	Male	Yes	382,581	579.63	29,854	GR	B.1.1.25	20B
DU-BCSIR-28	25	SRR14484356	EPI_ISL_2045661	40	Male	No	210,507	161.91	29,833	G	B.1	20A
DU-BCSIR-29	21	SRR14484355	EPI_ISL_2046273	43	Female	Yes	274,705	204.34	29,836	GH	B.1.351	Beta.V2
DU-BCSIR-30	25	SRR14484354	EPI_ISL_2046750	39	Male	Yes	283,873	162.58	29,838	GH	B.1.351	Beta.V2
DU-BCSIR-31	24	SRR14484353	EPI_ISL_2047053	25	Female	No	76,795	7548.04	29,836	GH	B.1.351	Beta.V2
BCSIR-01	24	SRR14484363	EPI_ISL_1673680	41	Female	Yes	262,584	78.21	29,836	GH	B.1.351	Beta.V2
BCSIR-02	25	SRR14484362	EPI_ISL_1673673	43	Male	Yes	442,244	931.89	29,836	GH	B.1.351	Beta.V2
BCSIR-661	NA	SRR14484352	NA	23	Male	No	1,230,812	NA	NA	NA	NA	NA
BCSIR-667	NA	SRR14484351	NA	27	Male ^a	No	574,958	NA	NA	NA	NA	NA

Abbreviation: RT-PCR, reverse transcriptase polymerase chain reaction.
^aLineage confirmation with probability 1.

2.7 | Bioanalysis of metagenomics data covering microbial diversity and antimicrobial profiling

IDseq—an open-source cloud-based pipeline has been used to assign taxonomy and AMR gene. This pipeline used quality control, host filtering, assembly-based alignment, and taxonomic reporting aligning to NCBI nt (nucleotide database). Trimmomatic²² and Bowtie2³¹ have been implemented for quality filtering, short-read alignment, and host filtering. For the AMR gene, threshold coverage was set ≥ 10 in the IDseq pipeline where ARG-Annot-2 database³² has been implemented. IDseq uses a third-party tool called SRST2³³ for AMR identification.

2.8 | Statistical analysis

Alpha diversity (diversity within samples) was estimated using chao1, Shannon, and Simson matrix, and the significance was tested using the Kruskal–Wallis rank test among vaccinated, nonvaccinated, and male, female samples. Principal coordinate analysis based on the Bray–Curtis distance method (beta-diversity) and permutational multivariate analysis of variance (PERMANOVA) was performed with 999 permutations to estimate a *p*-value for differences between vaccinated, nonvaccinated, and male and female samples. Phyloseq v3.12,³⁴ an R package, was used for microbial diversity analysis. Mean values of different groups were compared in IBM SPSS Statistics 20 software and *p*-values less than 0.05 at a 95% confidence level were considered as significant.

3 | RESULTS

3.1 | Detection of SARS-CoV-2 by RT-PCR and mNGS

The nasopharyngeal (NP) samples were initially screened for SARS-CoV-2 infection by RT-PCR based on cycle threshold (C_T) values. C_T values of the positive samples ($n = 17$) ranged between 21 and 26 (average $C_T = 23.88$). Negative samples ($n = 2$) showed no amplification curve in the PCR (Table 1). Average C_T values of vaccinated and nonvaccinated patients' samples were 24.6 and 23.43, respectively. But no significant difference was found between C_T values of vaccinated and nonvaccinated ($p = 0.06$), and between male and female ($p = 0.298$) patients. However, C_T values were negatively correlated ($r = -0.040$) with patients' age. Noteworthy, regardless of vaccination status, gender or age, no patients needed hospitalization in the sample set of this study. Furthermore, for the confirmation of SARS-CoV-2 infection, the metagenomic sequences were aligned against the SARS-CoV-2 reference genome (GenBank accession no. MN996528.1). All the positive samples (tested by RT-PCR) aligned with the reference genome, whereas metagenomic sequences of the negative samples did not align with the reference (Table 1).

3.2 | Mutational profile and selection analysis of the Bangladeshi SARS-CoV-2 genomes

To better characterize the genomes recovered with greater impact in the second wave of the epidemic in Bangladesh, we focus on describing circulating SARS-CoV-2 variants in vaccinated and non-vaccinated patients. Consensus genome sequences were generated from the metagenomic raw sequences of the samples that were tested COVID-19 positive by PCR, using the online platform EzCOVID19 (<https://www.ezbiocloud.net/tools/sc2>). Two deletion sites expanding varied nucleotides (nt) in S (position 717–742) and NSP6 (position 715–733) genes were evident in the area-wise coverage observation of the sequences using snippy. After error correction, consensus genomes have been generated keeping the deletions in consideration. Compared with the reference strain hCoV-19/Wuhan/WIV04 (GenBank accession no. MN996528.1), two consensus 9-nt inframe deletions were confirmed in S (nt position 719–727) and NSP6 (nt position 316–324) of the SARS-CoV-2 genomes originated from both vaccinated and nonvaccinated patients except the isolate DU-BCSIR-27 that was originated from a vaccinated male patient aged 65 years (Figure 1, Table 1). Mutational analysis revealed a similar mutational pattern in vaccinated and nonvaccinated genomes. Compared with the reference strain (GenBank accession no. MN996528.1), numerous amino acid (aa) substitutions were found in ORF1ab, S, ORF3a, E, and N proteins of the genomes ($n = 17$) generated in this study (Table 2). Among the ORF1ab proteins, NSP3 underwent comparatively higher aa substitutions while NSP4 and NSP16 of the isolates had almost no substitution except DU-BCSIR-24 (S34F in NSP4) and DU-BCSIR-27 (M270I in NSP16). Other ORF1ab proteins exhibited some limited aa substitutions such as NSP2 (T85I), NSP5 (K90R), NSP12 (P323L), and NSP13 (I151V) (Table 2). The RBD region of S protein of all the isolates in our sampling set also harbored common substitutions including K417N, E484K, and N501Y that signify the Beta, V2 (South African variant; B.1.351) implicating the higher transmissibility and infectivity of the variant.³⁵ Moreover, D614G, D215G, D80A, A67V, L18F, and A701V substitutions were commonly found in the non-RBD region of Spike proteins of both vaccinated and nonvaccinated samples (Table 2).

Furthermore, whole-genome-based mutation density and selection analyses were performed with a total of 2174 (17 sequences from this study and the rest downloaded from the GISAID database) complete and high-quality Bangladeshi SARS-CoV-2 genomes as of December 13, 2021. Among the proteins, ORF3a gene exhibited comparatively higher mutation density (46.98%) followed by NSP6 (44.02%), N (35.32%), and ORF8 (32.79%) (Figure 2). The mutational density of the S gene was 21.4% reflecting a high proportion of the nt positions undergoing frequent mutational evolution.

SLAC algorithm predicted ORF7b and ORF3a genes to have dN/dS ratio 1.77 and 1.24, respectively, indicating the positive selection of the genes. presence of higher number of missense than synonymous mutations in the genes (Table 3). We found ORF6, ORF7a, and N genes were genes to have dN/dS ratio of <1 , indicating negatively

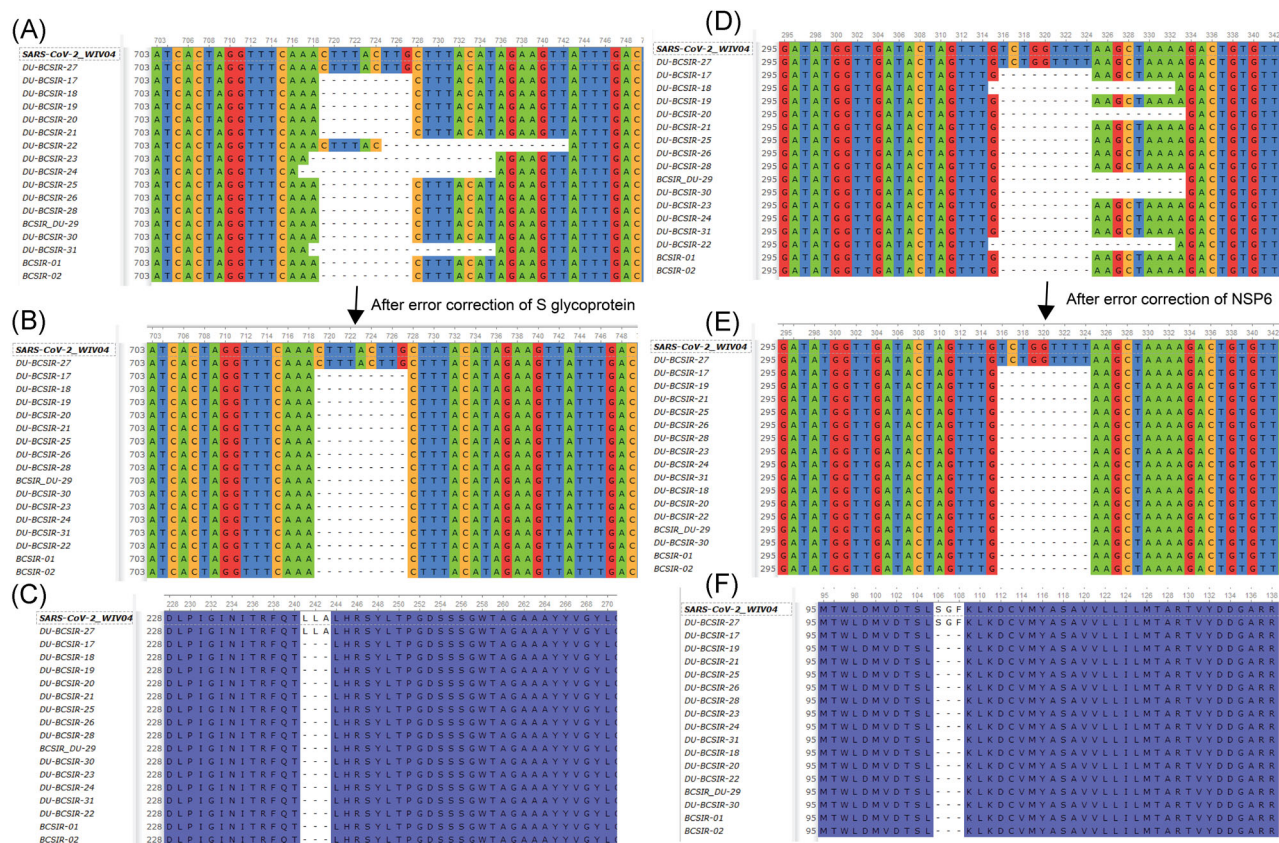


FIGURE 1 SARS-CoV-2 deletion discovered in spike glycoprotein and orf1ab (NSP6). (A) Deletion in S glycoprotein found after assembly where a few abnormal deletions was observed in four samples (DU-BCSIR-22, DU-BCSIR-23, DU-BCSIR-24, and DU-BCSIR-31). (B) After error correction (described in Section 2), a uniform nine nucleotide deletion in the N-terminal part of the RBD region of S protein and (C) resulted in three amino acid losses (LLA). (D) Abnormal deletion was also witnessed in five samples (DU-BCSIR-18, DU-BCSIR-20, DU-BCSIR-22, DU-BCSIR-29, and DU-BCSIR-30). (E) Here also a significant change in deletion after error correction and a uniform nine nucleotide deletion from nonstructural protein 6 (NSP6) of orf1a and (F) resulted in three amino acid losses (SGF) from NSP6

selection of the genes with time (Table 3). On the other hand, nine position (coding position (CP) = 19,67,95,142,143,452,681,701,950) and 20 coding positions (CP = 53,206,290,296,413,432,475,488, 489,508,543,586,683,691,764,824,1061,1062,1149,1237) (Table 3) were selected as positive and negative, respectively, showing overall negative selection (dN/dS ratio = 0.621) of S gene. The other protein coding genes were under negative selection pressure (Table 3).

3.3 | Phylodynamics and variant distribution of circulating SARS-CoV-2 in Bangladesh

To reveal the variants associated with the second wave of the COVID-19 pandemic among vaccinated and nonvaccinated patients in Bangladesh, Nextstrain phylogenetic and variant calling analyses were performed with the SARS-CoV-2 sequences. The Nextstrain Phylogenetic tree shows the clade distribution of the SARS-CoV-2 genomes originated from the vaccinated and nonvaccinated patients (Figure 3A). Most of the isolates (15/17, 88.24%), regardless of their origin vaccinated or nonvaccinated, clustered into the Beta variant (20H (Beta, V2); Pangoline B.1.351). The rest of the two isolates, DU-BCSIR-28 (non-vaccinated) and

DU-BCSIR-27 (vaccinated) belonged to the clades 20A (Pangoline B.1) and 20B (B.1.1.25), respectively (Table 1, Figure 3A). The Nextstrain Phylogenetic tree with all the Bangladeshi isolates represents the overall clade/lineage distribution of SARS-CoV-2 throughout the epidemic in Bangladesh (Figure 3B). Majority of the Bangladesh isolates clustered into the clade 20B (847/2174, 38.96%), followed by the clades 21J (Delta) (458/2174, 21.07%), 21A (Delta) (387/2174, 17.80%), 20H (Beta, V2) (249/2174, 11.45%), 20A (118/2174, 5.42%) and other 5.29% (115/2174) belonged to 8 variants (20I (Alpha, V1), 21I (Delta), 21D (Eta), 19B, 19A, 20C, 20F, 20J (Gamma)) (Figure 3B).

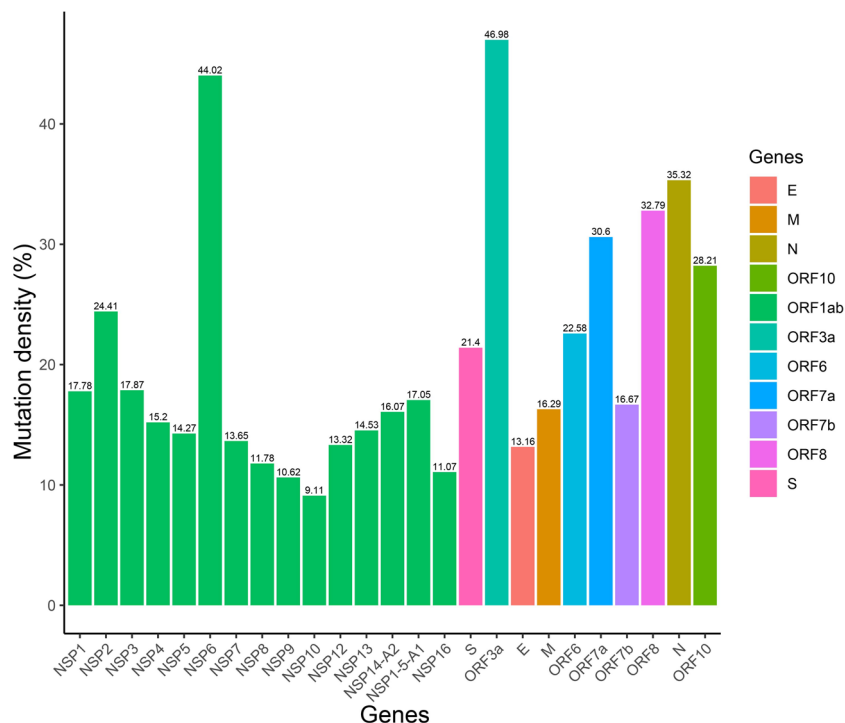
The monthly distribution of the variants in Figure 4 reveals that the first wave of the pandemic was dominated by clade 20B (April 2020 to February 2021, prevalence within this period 832/987, 84.30%) that continues till March 2021 (prevalence 11/126, 8.73%). The 20I (Alpha, V1) variant (UK variant) and the 20H (Beta, V2) variant (South African variant) were introduced in January 2021 in Bangladesh with a prevalence of 14.81% (12/81) and 2.47% (2/81), respectively. In February 2021, the Alpha, V1 variants cut 38.10% (16/42) of the deposited sequences, while 20B clade decreased to 42.85% (18/42) of the reported genomes. Dramatically, the Beta, V2 variant spread over and dominated (91/126, 72.22%) the second

TABLE 2 SARS-CoV-2 genome (sequenced) with their mutational profile compared with Wuhan strain (GenBank accession no. MN996528)

Gene Sample ID	ORF1ab NSP2	S										ORF3a	E	N
		NSP3	NSP4	NSP5	NSP6	NSP12	NSP13	NSP16	RBD region		Other than RBD region			
DU-BCSIR-17	T85I	P1228L	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), E484K	K417N, D614G,D215G,D80A, A701V,	Q57H,S171L	P71L	T205I	
DU-BCSIR-18	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), E484K,	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-19	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-20	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-21	T85I	N1778S,P1228L, P389L,K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-22	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-23	T85I	N1778S,P1228L ,K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, A67V,L18F,A701V	Q57H,A59T, S171L	P71L	T205I	
DU-BCSIR-24	T85I	N1778S,P1228L, K837N	S34F	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, A67V,L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-25	T85I	N1778S,P1228L, P389L,K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-26	T85I	N1778S,P1228L, P389L,K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-27	I120F	K412N	—	—	—	P323L	—	M270I	L452R	P681R,D614G	—	G204R, R203K		
DU-BCSIR-28	—	N1778S	—	—	Del106-108(SGL)	P323L	—	—	Del242-244(LAL)	D614G,D80A	S171L	P71L	—	
DU-BCSIR-29	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	
DU-BCSIR-30	T85I	P1228L,K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L		
DU-BCSIR-31	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, A67V,L18F,A701V	Q57H,A59T, S171L	P71L	T205I	
BCSIR-01	T85I	N1778S,P1228L, K837N	—	K90R	Del106-108(SGL)	P323L	—	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D80A,L18F, A701V	Q57H	P71L	T205I	
BCSIR-02	G481S,T85I, S138L	N1778S,P1228L, K837N,S126L	—	K90R	Del106-108(SGL)	P323L, T739I	I151V	—	Del242-244(LAL), K417N, E484K, N501Y	D614G,D215G,D80A, L18F,A701V	Q57H,S171L	P71L	T205I	

Abbreviations: LAL, leucine, alanine; leucine; SGL, serine, glycine, leucine.

FIGURE 2 The mutation density of the SARS-CoV-2 genome's coding region has been reported from Bangladesh. Most mutations occurred in ORF3a (46.98%) and minimum mutations occurred in NSP10 (9.11%). Here x-axis represents the gene name and the y-axis represents the DNA mutation density (total mutation/length (bp) of the gene)



wave of the epidemic in March 2021 and the dominance continues through April 2021 (93/111, 83.78%), and May 2021 (46/104, 44.23%). However, Delta variants (21J and 21A) were the most widespread variants (838/947, 88.49%) from May 2021 to October 2021, which were first found in Bangladesh in May 2021³⁶ (Figure 4, Supporting Information Data 1).

3.4 | Assessing co-infections through metagenomics approach

Metagenomics was used to assess the presence of possible coinfections with bacteria, viruses (other than SARS-CoV-2), fungi, and/or parasitic microorganisms. Possible coinfections of the SARS-CoV-2 positive and control samples were mentioned in Table 4. Of the 17 COVID-19-positive samples from both vaccinated and nonvaccinated groups of patients, 16 (94.12%) of COVID-19 cases were coinfecting by at least one clinically relevant microorganism. Among these, 1 (5.88%) was infected with only one bacterium among the vaccinated group of patients and the remaining 15 (88.245%) were coinfecting with two or more pathogenic bacteria regardless of the vaccination status. High rates of common, pathogenic/opportunistic bacterial coinfections were detected, which included *Streptococcus agalactiae* ($n = 9$; 52.94%), *Neisseria meningitidis* ($n = 10$; 58.82%), *Elizabethkingia anophelis* ($n = 10$; 58.82%), *Stenotrophomonas maltophilia* ($n = 10$, 58.82%), *Klebsiella pneumoniae* ($n = 12$, 70.59%) and *Pseudomonas plecoglossicida* ($n = 5$, 29.41%) (Table 4). In our SARS-CoV-2-negative samples, *S. agalactiae*, *Neisseria mucosa*, *Neisseria meningitidis*, *S. maltophilia* and *K. pneumoniae* were also identified with a very low abundance, while no fungal, viral other than SARS-CoV-2 or protists coinfections were detected (Table 4). However, we did not find any

significant correlation among associated factors including vaccination status, age or sex of the patients, diversity or abundance of the co-infected organisms/pathogens, and the abundance of SARS-CoV-2.

3.5 | Microbiome diversity and composition

We investigated the strain-level microbial community and relative abundances in the healthy and COVID-19 patient samples. The reads generated from metagenomic sequences (MS) were mapped to a total of 1149 bacteria, 189 Eukaryota, and 40 viruses through the IDSeq microbiome analysis pipeline (Supplementary Data 1). We did not, however, find any significant differences in the alpha and beta diversities among the microbial communities between or across the samples (Figure 5A, B, respectively). Alpha diversity showed no significance in Kruskal–Wallis rank-sum test between sex ($p = 0.2234$) and among vaccination, nonvaccination, and control ($p = 0.6556$) with Shannon alpha diversity measure. Simpson alpha diversity measure showed no significance in Kruskal–Wallis rank-sum test between sex ($p = 0.3029$) and among vaccination, nonvaccination, and control ($p = 0.4367$). Similarly, no significance was shown in Kruskal–Wallis rank-sum test between sex ($p = 1$) and among vaccination, nonvaccination, and control ($p = 0.92$) with Chao1 alpha diversity measure. For beta diversity, the PERMANOVA test significance between males and females was nonsignificant ($p = 0.303$). Also, beta diversity PERMANOVA test significance among vaccination, nonvaccination, and control was nonsignificant ($p = 0.081$). Though the previous study found significant differences in microbiome composition between COVID-19 and non-COVID-19 samples³⁷ in Bangladesh. Investigation with large samples both healthy and clinical samples are recommended for conclusive remarks.

TABLE 3 SARS-CoV-2 codon with their number of the synonymous and nonsynonymous mutational profile, dN/dS ratio, and direction of selection

Gene	Gene length	Total no. of mutations	No. of nonsynonymous mutations	No. of synonymous mutations	Mutation density (%)	dN/dS ratio	Positively selected codon position ($p \leq 0.1$)	Negatively selected Codon Position (CP) ($p \leq 0.1$)
NSP1	540	96	40	56	17.78	0.493	—	1 (CP = 9)
NSP2	1914		280	168	24.41	0.815	1 (CP = 120)	5 (CP = 22,132,161,585,629)
NSP3	5835	1043	597	446	17.87	0.550	3 (CP = 488,822,1228)	26 (CP = 12,39,106,207,311,339,356,414,758, 807,849,955,1050,1142,1197,1234,1283, 1354,1401,1410,1508,1558,1705,1739,1869)
NSP4	1500	228	116	112	15.2	0.459	1 (CP = 446)	4 (CP = 121,326,335,337)
NSP5	918	131	64	67	14.27	0.411	—	5 (CP = 28,105,187,2229,305)
NSP6	870	383	92	291	44.02	0.616	2 (CP = 37,149)	3 (CP = 168,227,270)
NSP7	249	34	13	21	13.65	0.278	—	—
NSP8	594	70	34	36	11.78	0.368	—	—
NSP9	339	36	15	21	10.62	0.364	—	1 (CP = 87)
NSP10	417	38	19	19	9.11	0.411	—	1 (CP = 108)
NSP12	2793	372	184	188	13.32	0.450	1 (CP = 670)	10 (CP = 144,259,273,293,329,599,712,714,765, 839,866)
NSP13	1803	262	146	116	14.53	0.521	1 (CP = 77)	6 (CP = 19,217,245,294,306,348)
NSP14-A2	1581	254	144	110	16.07	0.614	1 (CP = 394)	5 (CP = 93,252,326,377,517)
NSP1-5-A1	1038	177	124	53	17.05	0.746	—	2 (CP = 73,176)
NSP16	894	99	56	43	11.07	0.411	—	1 (CP = 133)
S	3822	818	527	291	21.40	0.621	9 (CP = 19,67,95,142, 143,452,681,701,950)	20 (CP = 53,206,290,296,413,432,475,488, 489,508,543,586,683,691,764,824,1061, 1062,1149,1237)
ORF3a	828	389	292	97	46.98	1.24	2 (CP = 26,151)	8 (CP = 2,43,65,113,204,207,212,273)
E	228	30	17	13	13.16	0.518	—	—
M	669	109	46	63	16.29	0.355	1 (CP = 82)	—
ORF6	186	42	28	14	22.58	0.730	—	1 (CP = 23)
ORF7a	366	112	79	33	30.60	0.928	—	2 (CP = 40,52)
ORF7b	132	22	17	5	16.67	1.77	—	—
ORF8	366	120	84	36	32.79	0.846	—	2 (CP = 28,120)

TABLE 3 (Continued)

Gene	Gene length	Total no. of mutations	No. of nonsynonymous mutations	No. of synonymous mutations	Mutation density (%)	dN/dS ratio	Positively selected codon position ($p \leq 0.1$)	Negatively selected Codon Position (CP) ($p \leq 0.1$)
N	1260	445	261	184	35.32	0.662	1 (CP = 63)	23 (CP = 16,35,41,86,110,115,157,172,192,228, 268,274,298,302,309,315,318,329,333, 341,360,362,367)
ORF10	117	33	23	10	28.21	0.916	—	—

The strain-level signature of the microbiome demonstrated that most of the species identified in each sample were represented by multiple strains (Supplementary Data 1), and we identified the top 70 strains according to their relative abundance (Figure 6). The predominant bacterial phyla we found among the samples were Proteobacteria, Firmicutes, Actinobacteria, Bacteroidetes, and Basidiomycota (contributing to >95.0% of the total sequences). Basidiomycota, Ascomycota, Platyhelminthes, Streptophyta, and Apicomplexa were predominant eukaryotic phyla that reflect possible infection with fungi and worms both in healthy and clinical samples, not significantly distinguishable. Vaccinated samples contained the viruses from Siphoviridae, Picornaviridae, Inoviridae, Adenoviridae, Siphoviridae, Herpesviridae, and Parvoviridae family and non-vaccinated samples had viruses of Siphoviridae, Adenoviridae, and Podoviridae, whereas the control data harbored only one virus namely Streptococcus phage Javan269.

Among non-COVID-19-control, nonvaccinated, and vaccinated people, the high abundance species ($\geq 0.05\%$) number was 43, 145, and 77, respectively. Among them, five species were shared among all groups. Forty-three unique species were observed in vaccinated people whereas 109 and 26 unique species were observed in non-vaccinated and non-COVID-19-control samples (Figure 7B, Table S1). The top five microbial species in vaccinated samples were *Enterobacter cloacae* (32.81%), *Brevundimonas diminuta* (10.49%), *S. maltophilia* (9.38%), uncultured bacterium (7.69%) and *Cutibacterium acnes* (7.26%), whereas *E. cloacae* (13.11%), uncultured bacterium (12.29%), *B. diminuta* (10.20%), *Sphingobacterium multivorum* (8.41%) and *S. maltophilia* (5.51%) were noticed in nonvaccinated samples. In non-COVID-19-control samples, *Streptococcus salivarius* (15.94%), *Rothia mucilaginosa* (15.45%), *Streptococcus sp.* oral taxon 431 (13.14%), *Streptococcus mitis* (9.86%) and *Veillonella atypica* (9.35%) were detected (Supplementary Data 1). *S. maltophilia*, *B. diminuta*, *E. cloacae*, and uncultured bacterium are common between vaccinated and nonvaccinated samples. When high abundance species ($\geq 0.05\%$) in male, female people were considered, there were 64 species found in females whereas only 28 species found in males. Among the microbial species, 16 are common to both, 12 are found in only males, and 48 species are found in only females (Figure 7C, Table S1). The top five high abundance microbes found in males were *E. cloacae* (15.67%), *B. diminuta* (8.09%), *C. acnes* (5.40%), *S. salivarius* (5.19%) and *S. maltophilia* (5.03%). In females, among top five species, there were also present *E. cloacae* (33.16%), *B. diminuta* (11.45%), *S. maltophilia* (10.43%) which are also present in males, and in addition uncultured bacterium (19.78%) and *V. atypica* (3.58%) were present in notable abundance (Supplementary Data 1).

3.6 | Antibiotics resistance gene families in healthy control and COVID-19 patients

We further investigated the correlation and/or influence of the AMR between vaccinated and nonvaccinated COVID-19 patients' diseases processes present in the microbiomes of these four sample

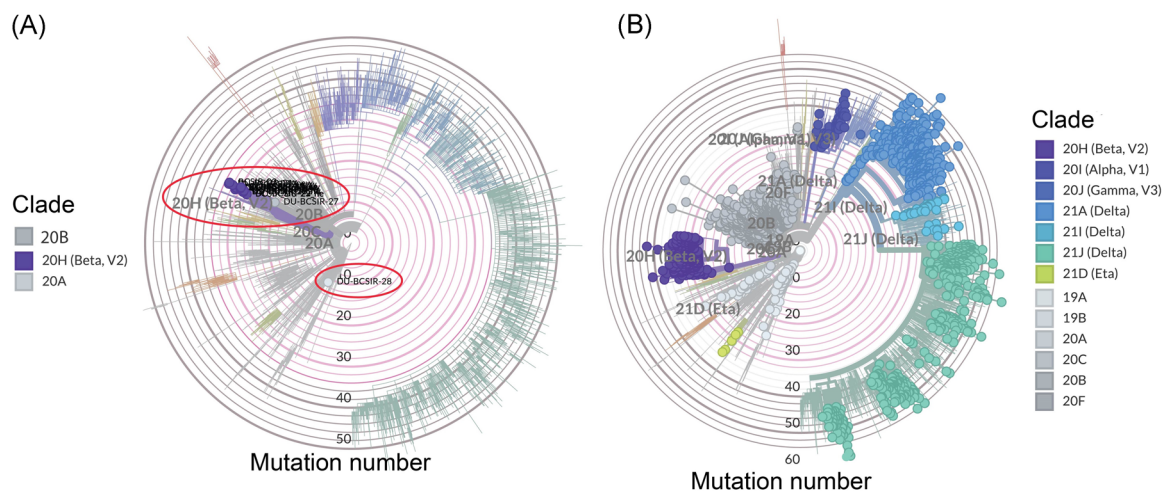


FIGURE 3 Phylogenetic tree of (A) 17 sequenced SARS-CoV-2 genomes (marked with a red circle) and (B) total 2174 Bangladesh sequences available at GISAID. The variants are represented by a color code

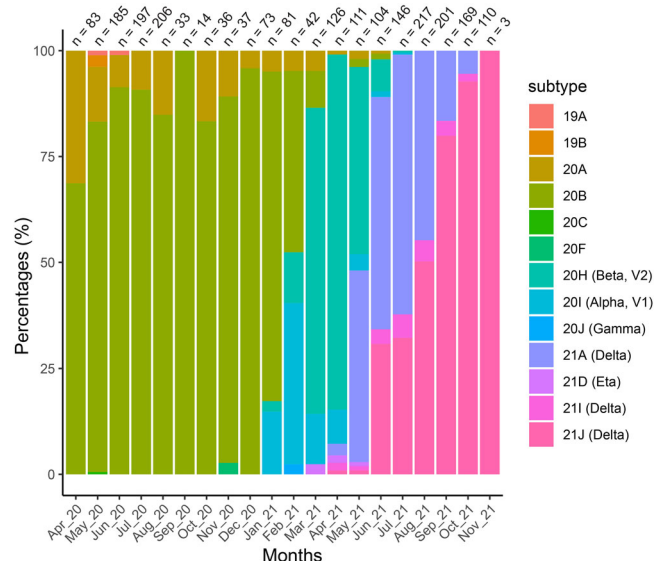


FIGURE 4 Dynamics of SARS-CoV-2 variants (Nextclade clade) from March 2020 to November 2021 in Bangladesh

categories using the IDseq pipeline. In our current metagenomic data set, there was a broad variation in ARGs diversity and composition (Supplementary Data 2). IDseq pipeline identified a total of 34 ARGs belonging to different resistance mechanisms (Supplementary Data 2). Most of the ARGs involve efflux mechanism ($n = 8$) followed by class A beta-lactamases and tetracycline resistance ribosomal protection proteins both of which count four resistance genes (Figure 7). Among all the resistance genes, maximum reads ($n = 36$) mapped against *cfxA*, *tet-37*, and *ermB* genes in the samples DU-BCSIR-31, DU-BCSIR-17, and DU-BCSIR-28, respectively. Five AMR genes (*aadA*, *aph3''la*, *dfrA*, *LAP*, and *tetX*) were observed in ($n = 9$, 47.37%) samples followed by *cmlA* ($n = 8$, 42.11%), *qnr-S* ($n = 7$, 36.84%), and *sul3*, *tetA* ($n = 6$, 31.58%). The rest of the AMR

represents less than in 21% samples. Noteworthy, compared with the healthy samples, higher ARGs were found in COVID-19 samples (Figure 7). A total of 20 AMR genes were observed in nonvaccinated SARS-CoV-2-positive patients whereas the number increased to 22 in vaccinated COVID-19-positive patients. In non-COVID-19 samples, there were only eight AMR genes observed. There are 12 common genes (*LAP*, *dfrA*, *sul3*, *tetX*, *tetA*, *sul2*, *qnr-S*, *aadA*, *strB*, *aph3''la*, *mefB*, *cmlA*) between nonvaccinated and vaccinated whereas non-COVID-19-control and nonvaccinated contained three common AMR genes (*mefA*, *cfxA*, *ermX*). *msrD* is common between non-COVID-19-control and vaccinated samples. Nine genes (*ereA*, *aphA2*, *strA*, *catBx*, *tetG*, *tet-37*, *aac3-IIa*, *qxB*, *mphE*) are unique to vaccinated samples followed by five genes (*floR*, *AER-1*, *ermB*, *tetW*, *aac3-Ib*) to nonvaccinated samples and four (*penA*, *tetQ*, *PBP1b*, *tetM*) for non-COVID-19-control (Figure 8A). Among male and female 16 AMR (*LAP*, *msrD*, *dfrA*, *sul3*, *tetX*, *tetA*, *strA*, *sul2*, *catBx*, *cfxA*, *tetG*, *qnr-S*, *aadA*, *aph3''la*, *mefB*, *cmlA*) are common to both, 16 (*floR*, *aphA2*, *penA*, *AER-1*, *mefA*, *ermB*, *tetQ*, *strB*, *PBP1b*, *tet-37*, *tetW*, *qxB*, *mphE*, *tetM*, *ermX*, *aac3-Ib*) only found in males and 2 (*ereA*, *aac3-IIa*) found in females (Figure 8B).

4 | DISCUSSION

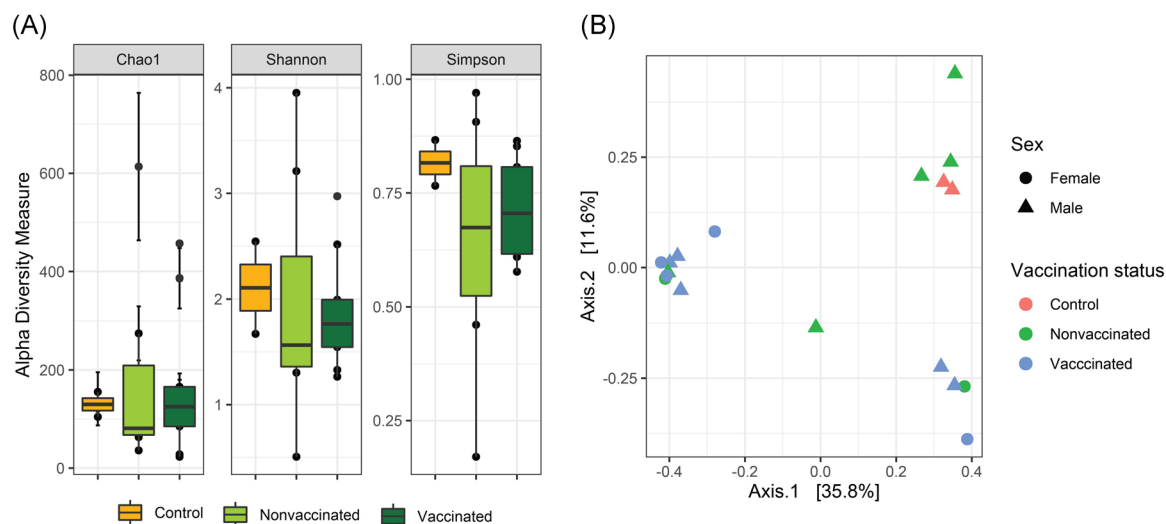
In the present study, we characterized the genomic and metagenomic features of the NPS from vaccinated (Oxford-AstraZeneca) and nonvaccinated COVID-19 patients using next-generation sequencing-based metagenomic and whole-genomic approaches. In the preliminary screening of the samples by RT-PCR, we found comparatively higher C_T values in the vaccinated patients (average $C_T = 24.6$) than the nonvaccinated patients (average $C_T = 23.43$), but statistically insignificant ($p = 0.06$). C_T values were also found regardless of the patients' gender either male or female ($p = 0.298$). However, a statistically negative correlation was found between the C_T values

TABLE 4 Coinfections in SARS-CoV-2-positive patients. Number represents the relative abundance (%)

Sample ID	<i>Streptococcus agalactiae</i>	<i>Neisseria meningitidis</i>	<i>Elizabethkingia anophelis</i>	<i>Stenotrophomonas maltophilia</i>	<i>Klebsiella pneumoniae</i>	<i>Pseudomonas plecoglossicida</i>
BCSIR-01	1.52	0	0	0	0	0
BCSIR-02	0	0	0	0	0	0
BCSIR-661 ^a	0	1.64	0	0	0	0
BCSIR-667 ^a	0.037	0.097	0	0.001	0.005	0
DU-BCSIR-17	0.13	0.19	0.12	0.91	0.88	0
DU-BCSIR-18	1.45	13.49	0	0	0	0
DU-BCSIR-19	0.2	2.39	0	1.2	0.4	0
DU-BCSIR-20	0	0.01	1.17	9.25	0.39	2.47
DU-BCSIR-21	0	0.71	5.65	10.23	1.3	0
DU-BCSIR-22	0	0	0.35	0	0.5	0
DU-BCSIR-23	0	0	1.79	16.82	0.35	2.26
DU-BCSIR-24	1.4	0	0	0	0	0
DU-BCSIR-25	0.01	0.06	0.22	21.72	0.36	2.61
DU-BCSIR-26	0	0.01	0.24	14.34	0.4	0
DU-BCSIR-27	0.01	0.08	0.08	8.65	0.34	0
DU-BCSIR-28	0.03	0.01	0	0	0.001	0
DU-BCSIR-29	0	0.01	0.15	22.16	0.23	2.48
DU-BCSIR-30	0.01	0	2.67	18.88	2.07	0.08
DU-BCSIR-31	6.22	0	0	0	0	0

Note: Number represents the relative abundance (%).

^aRT-PCR-negative samples (Control).

**FIGURE 5** Alpha and beta diversity measure of mRNA microbiome taxonomy data. (A) Chao1, Shannon, and Simpson alpha diversity measure among vaccinated, unvaccinated, and control samples (unvaccinated and COVID-19 negative). (B) Principal component analysis with Bray–Curtis distance matrix

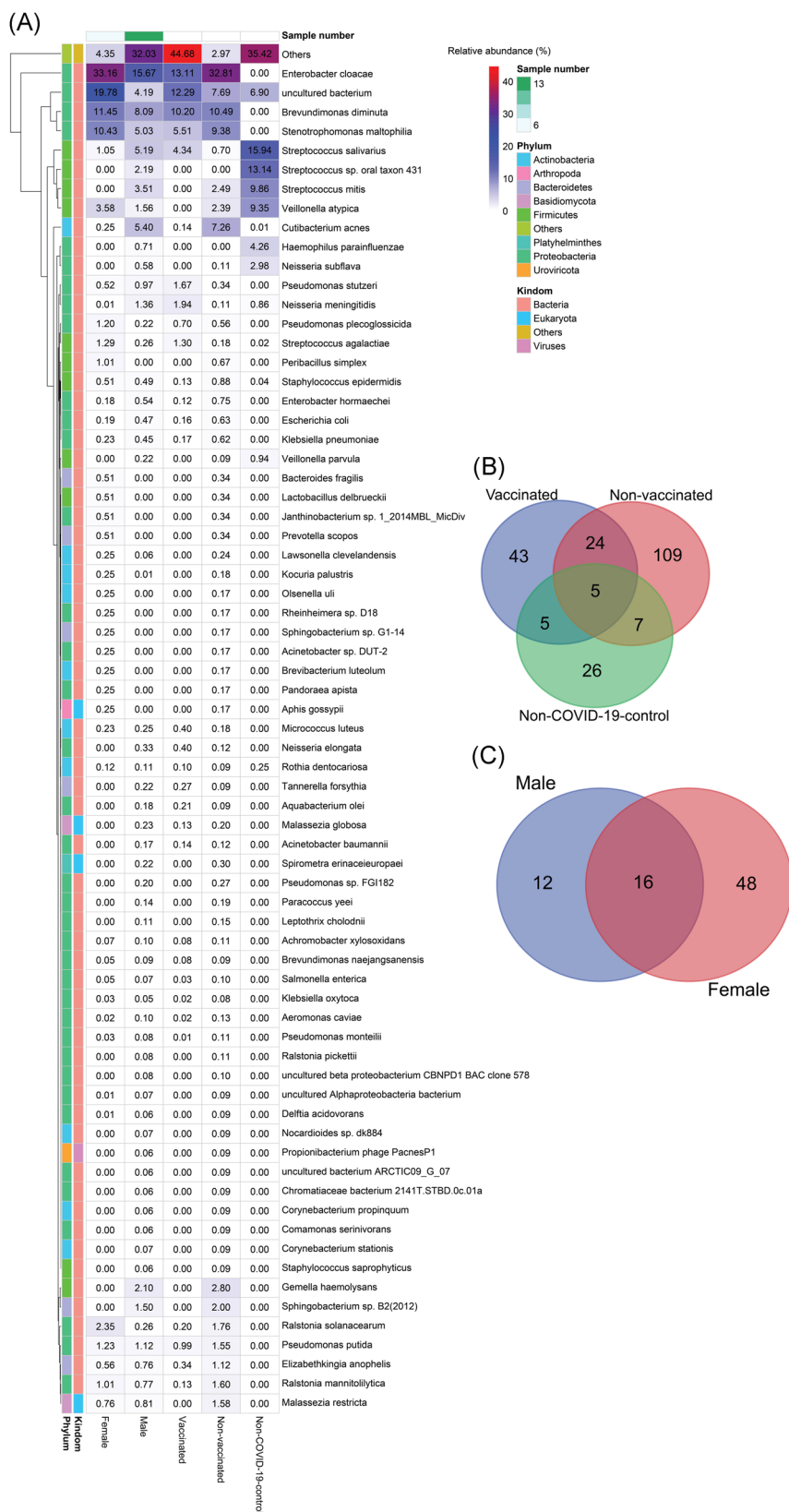


FIGURE 6 (A) Top 71 microbial species with their relative abundance (%) from mRNA metagenome analysis. The number represents the percentages. To maintain uniformity, the SARS-CoV-2 genome relative abundance is not considered here. High abundance species ($\geq 0.05\%$) in male, female, non-COVID-19-control, nonvaccinated, and vaccinated samples were considered to observe the shared diversity. (B) Non-COVID-19-control, nonvaccinated and vaccinated people with their number of shared microbial species, and (C) represents the common and unique microbial number between males and females. The details can be found in Table S1

and the age of the patients ($r = -0.040$), indicating a likely higher viral load (lower C_T values) in older patients. These findings indicate that viral load in the NPS is independent of vaccination or gender, but inversely correlated with patients' age in the sample set of this study.

However, regardless of C_T values, gender, and age, all the patients of this study had a mild infection and did not need hospitalization. This mild clinical outcome could be related to the vaccination effect or previous infection (not confirmed) as high seroprevalence was

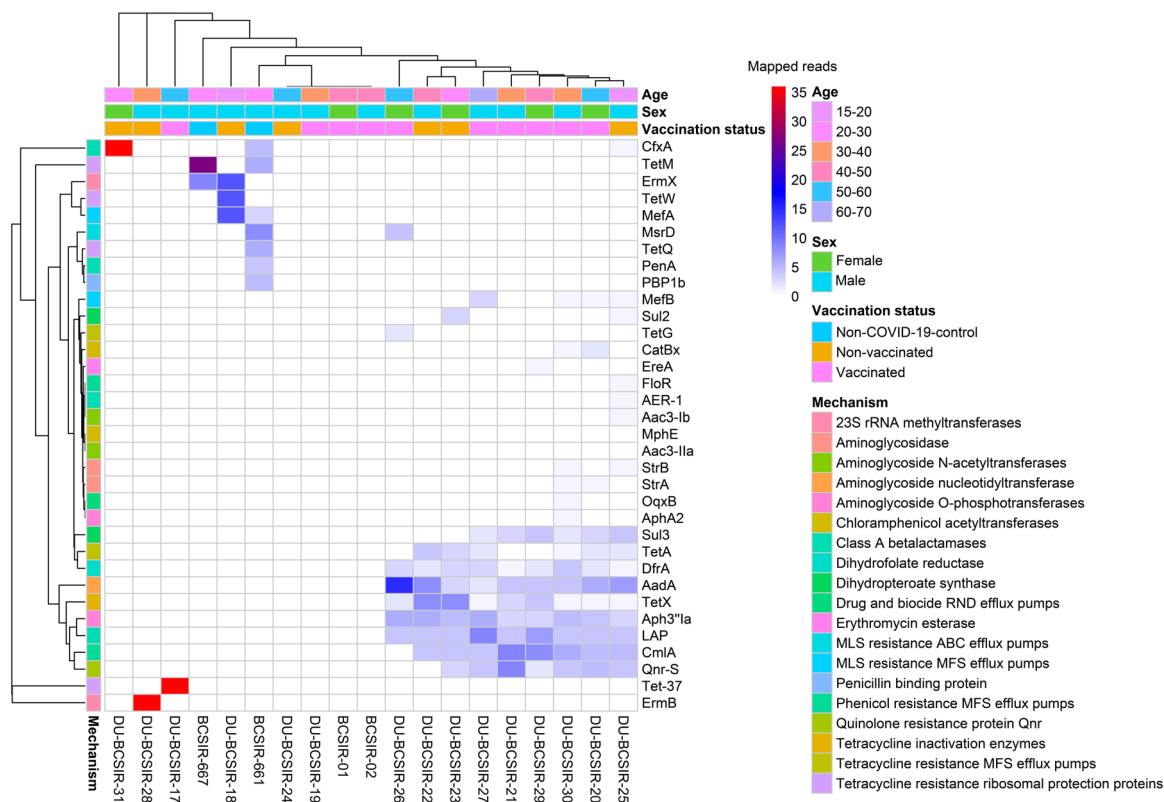


FIGURE 7 Antimicrobial resistance genes and their functions were discovered in 19 samples using mRNA metagenome analysis

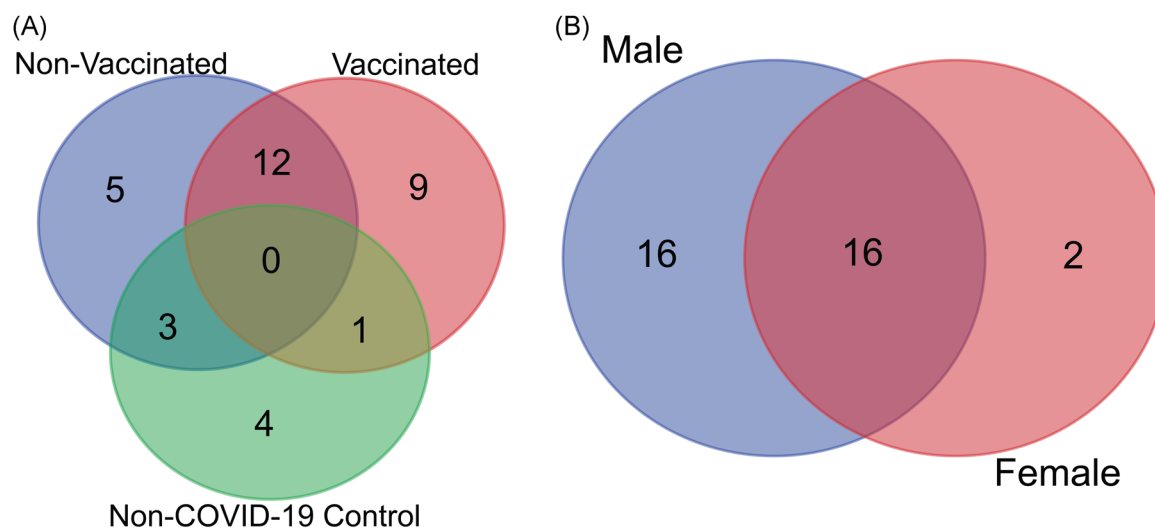


FIGURE 8 Antimicrobial resistance genes are depicted in a Venn diagram. (A) AMR sharing data among vaccinated, nonvaccinated, and non-COVID-19 control. (B) AMR sharing between male and female samples

reported in Dhaka, the capital of Bangladesh. Genomic and co-morbidity features of the indigenous patients might also affect the clinical outcome, which warrants further comprehensive study with a large sample number.^{38,39}

After the initial screening for SARS-CoV-2 by RT-PCR, both the positive ($n = 17$) and the negative samples ($n = 2$) were further

sequenced through metagenomic NGS and the raw sequences were aligned against the reference strain, hCoV-19/Wuhan/WIV04 (GenBank accession no. MN996528.1). All the RT-PCR-positive samples also aligned against the reference, while, expectedly, the negative samples did not align any reads against the reference genome. Thus, metagenomic sequencing could be a potential alternative to the

RT-PCR test for SARS-CoV-2 detection and importantly, for diagnosis of coinfection or microbiome composition in the COVID-19 patients.⁴⁰

Here, the complete genome sequences were generated from the metagenomic sequences and investigated for mutational spectra between vaccinated and nonvaccinated patients' samples. Error correction through area-wise coverage observation revealed two consensus 9-nt inframe deletions in S (nt position 719–727) and NSP6 (nt position 316–324) of the SARS-CoV-2 genomes originated from both vaccinated and nonvaccinated patients except the isolate DU-BCSIR-27 that originated from a vaccinated male patient aged 65 years. Deletion in antigenically important S proteins might affect the viral pathogenesis, transmissibility, infectivity, and disease severity.^{41,42} This finding warrants more careful area-wise coverage observation of the genomes while building a consensus genome of the virus for accurate complete genome sequences with consensus insertion or deletion.

Furthermore, the mutational analysis revealed a similar mutational pattern in vaccinated and nonvaccinated genomes. Compared with the reference strain (GenBank accession no. MN996528.1), a number of amino acid (aa) substitutions were found in ORF1ab, S, ORF3a, E, and N proteins of the genomes generated in this study. Among the ORF1ab proteins, NSP3 underwent comparatively higher aa substitutions than the other ORF1ab proteins that exhibited a limited aa substitution. The RBD region of S protein of all the isolates in our sampling set also harbored common substitutions including K417N, E484K, and N501Y that signify the Beta, V2 implicating the higher transmissibility and infectivity of the variant⁴³ that might have resulted in higher R_0 (3.5) number in the second wave in Bangladesh.⁴⁴ Moreover, whole-genome-based mutation density and selection analyses revealed ORF3a gene exhibited comparatively higher mutation density (46.98%), whereas the mutational density of the S gene was 21.4% reflecting a high proportion of the nt positions undergoing frequent mutational evolution. As a result of continuous mutations genes are seemingly going through natural selection. Selection analysis predicted ORF3b and ORF3a genes as positively selected (dN/dS ratio 1.77 and 1.24, respectively), indicating the presence of a higher number of missense than synonymous mutations in the genes. This finding indicates that ORF3a and ORF7b are rapidly evolving and are highly divergent. Previously, ORF6, ORF7a, and N genes were reported as positively selected genes with a dN/dS ratio of 17.8, 1.08, and 1.55, respectively,⁴⁵ while we found these genes to have a dN/dS ratio of <1, indicating negative selection of the genes with time. Noteworthy ORF6 protein is an accessory protein whose function is yet to be fully elucidated. Overall, the S gene selected as negative with 9 positively selected and 20 negatively selected coding positions reflecting that the surface glycoprotein (S) might approach toward positive selection pressure and evolve and thus warrants careful observation to trace the evolution of the SARS-CoV-2 S protein, critical target for vaccination and therapeutic intervention.^{46–49} Thus, this selection analysis revealed that some genes are under positive selection pressure indicating that the virus is fast-evolving presumably to evade the host cell's innate immunity, which should be taken into special consideration before vaccine development or other treatment strategies.

The monthly distribution of the variants in Figure 4 reveals that the first wave of the pandemic was dominated by clade 20B (April 2020 to February 2021), prevalence within this period that continues till February 2021. However, the introduction of the Alpha, V1 variant and Beta, V2 variant in January 2021 initiated the second wave of the epidemic in Bangladesh. The epidemic of February 2021 was dominated by Alpha, V1 with a gradual increase of the Beta, V2 variant that dramatically spread over the next month, March concurrent with the rapid resurgence and second wave of the COVID-19 epidemic in Bangladesh. After the first detection of Delta variations (Ref), the variants (21J, 21A) were predominant until August 2021, and then steadily decreased until November 2021 (Figure 4). So, the third wave of COVID-19 (May 2021 to September 2021) was responsible for Delta variants in Bangladesh, whereas the second wave (January 2021 to May 2021) was responsible for Beta, V2 variations. There have only been two reports of Omicron variants in Bangladesh until this report was submitted, which could represent a big concern and could be the cause of the third wave of COVID-19 in Bangladesh.⁵⁰ SARS-CoV-2 has been changing since its inception, resulting in Beta, Delta, and Omicron.^{41,51–55} As a large number of vaccinations⁵⁶ and treatments⁵⁶ are available, we may expect the virus's virulence to gradually decline.

The Nextstrain phylogenetic and variant calling analyses with the SARS-CoV-2 sequences revealed most of the sequences (15/17, 88.24%) generated in the present study belonged to the Beta, V2 variant, regardless of their originating samples either vaccinated or nonvaccinated. It identifies the dominant Beta variant associated with the second surge with a dramatic increase in the case and death reports in Bangladesh, corroborated with the previous reports.^{57–60} This resurgence might be correlated with the unrestricted movement of the mass population within the home or from abroad after the launch of the vaccine drive. Thus, it should be kept in mind that restricting movement and good health sanitation are important until the administration of the second dose vaccine to a sufficient population in the country.

Normal lockdown from April 5, 2021, and subsequent “strict lockdown” from April 14 which is going on. Second dose vaccination from April 8, administered to ~27% individuals, as of December 15, 2021.⁶¹ Noteworthy, the downward trends of the surge curve from mid-April could be associated with the re-intervention of the lockdown or with the administration of the second dose vaccine. The association of the Beta, V2 and Alpha, V1 variants with the recent resurgence of COVID-19 despite the high seroprevalence (ranged 45%–74%) of COVID-19 in Dhaka, Bangladesh reflects the possible antigenic escape of the recent variants from the previously reported variants or from the ChAdOx1 (Oxford–AstraZeneca) vaccine.⁶² These findings warrant urgent comprehensive studies focusing on the molecular as well as sero-surveillance for nonpharmaceutical interventions and vaccine policy.

Microbial coinfections have been considered as a principal causative agent of morbidity and mortality at the time of viral infections and also have been witnessed to become involved in COVID-19, although by a low number of COVID-19 cases.^{63,64} From our

metagenomics analysis, we did not find any significant microbial difference between vaccinated and nonvaccinated people and even no significance between males and females. A recent study found *Haemophilus influenzae*, *Moraxella catarrhalis*, human metapneumovirus, human alphaherpesvirus 1 coinfection in several samples with a significant abundance.⁴⁰ None of them were found at a significant level from our study but we found different bacterial coinfection like *S. agalactiae*, *N. meningitidis*, *E. anophelis*, *S. maltophilia*, *K. pneumoniae* and *P. plecoglossicida*. *S. agalactiae* ($n = 9$) was previously reported in nasal swab in SARS-CoV-2 positive patients.^{65,66} *N. meningitidis* ($n = 10$), a causative organism of meningococcal meningitis, was found with COVID-19 coinfection from a case report study⁶⁷ and it is also a coinfecting bacterial species which is also noticed in influenza coinfection.⁶⁸ *E. anophelis* ($n = 10$) is considered an emerging pathogen that implicates in cases of nosocomial pneumonia, sepsis, and neonatal meningitis⁶⁹ although not reported in any previously reported cases with COVID-19 coinfection. *S. maltophilia* ($n = 10$) has a synergic impact on the mortality of pneumonia patients.⁷⁰ *K. pneumonia* ($n = 12$) has previous reports of coinfection with COVID-19 in Italy.⁷¹ *P. plecoglossicida* ($n = 5$), a causative agent of bacterial hemorrhagic ascites in freshwater fish,⁷² is also found in COVID-19 patients although no previous report to human infection. These findings indicate likely coinfection with bacteria, fungi, or viruses in COVID-19, which may aggregate the clinical outcome of the disease. This might happen due to immune-suppression upon SARS-CoV-2 infection that leads to other viral coinfections or vice versa.^{73,74} However, further careful observation on microbial coinfection is warranted for understanding this phenomenon.

Besides microbial coinfection, it is noteworthy to indicate that antimicrobial-resistant bacterial agents may cause subsequent complications in COVID-19 cases where empiric use of antibiotics has been administered.⁷⁵ Broad-spectrum antimicrobial agents are prescribed at high rates to combat bacterial and fungal coinfection.⁷⁵ Azithromycin, ivermectin, and doxycycline were the most frequently used antibiotics in Bangladesh⁷⁶ and worldwide.⁷⁷ Methylation mediated by methylases encoded by *erm* genes (especially *ermA* and *ermB*) are responsible for Azithromycin⁷⁸ and *ermB* genes observed in a sample of our study. Tetracycline resistance MFS efflux pumps encoded by *tetA* and Tetracycline resistance ribosomal protection proteins encoded by *tetM* are responsible for doxycycline resistance. In this study, *tetA* ($n = 6$) and *tetM* ($n = 2$) antimicrobial genes are detected.

5 | CONCLUSION

Our study reveals genomic and metagenomic features of the SARS-CoV-2 associated with infections among vaccinated and nonvaccinated patients. The complete genome-based analysis identifies the Beta, V2 variant (South African variant) dominating the second surge of COVID-19 that occurred in early March 2021, including both the vaccinated and nonvaccinated patients in Bangladesh. Regardless of vaccination, no patient needed hospitalization and no significant

difference was found between the C_T values of vaccinated and nonvaccinated groups. In Bangladesh, the second wave of COVID-19 (January 2021 to May 2021) was responsible for Beta, V2 variations, while the third wave (May 2021 to September 2021) was responsible for Delta variants. In-depth metagenomic analyses reveal no significance in mutational profile, microbiome diversity, and AMR profile between vaccinated and nonvaccinated people and even no significance between males and females. Infection in vaccinated individuals with the South African variant warrants comprehensive serological analyses for vaccination policy and prophylactic strategy on an urgent basis.

Overall, this approach provides valuable insights on the composition of the respiratory microbiome from both vaccinated and nonvaccinated patients and pictures its relationship with coinfections and/or other organisms that may have a significant impact on SARS-CoV-2 disease progression and prognosis.

5.1 | Study limitation

Noteworthy, the genomes sequenced in this study represent a small proportion of confirmed SARS-CoV-2 cases in Bangladesh. Thus, these variants may not reflect the actual diversity of SARS-CoV-2 circulating in Bangladesh. However, this genomic surveillance may represent the genomic and metagenomics features of the variants associated with the infections among vaccinated and nonvaccinated patients during the second wave of the epidemic in the country.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

ETHICS STATEMENT

This study was approved by the Biological Faculty of Dhaka University institutional review board. We have taken the written consent of the participants.

AUTHOR CONTRIBUTIONS

Md. Mizanur Rahman and Md. Salim Khan developed the hypothesis, interpreted, drafted, and critically reviewed and finalized the manuscript. M. Shaminur Rahman, M. Rafiul Islam and Israt

Islam analyzed, interpreted, and drafted the manuscript. Otun Saha, Shahina Akter, Tanjian Akhtar Banu, Iffat Jahan, Md. Ahasan Habib, Barna Goswami, Latiful Bari, and Md. Abdul Malek critically reviewed the manuscript.

DATA AVAILABILITY STATEMENT

The sequences of 17 SARS-CoV-2 genomes were submitted to the GISAID database under the identifiers (Table 1). The raw reads were submitted to the NCBI SRA under BioProject accession number PRJNA728728.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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